BioThings Explorer: A query engine for a federated knowledge graph of biomedical APIs

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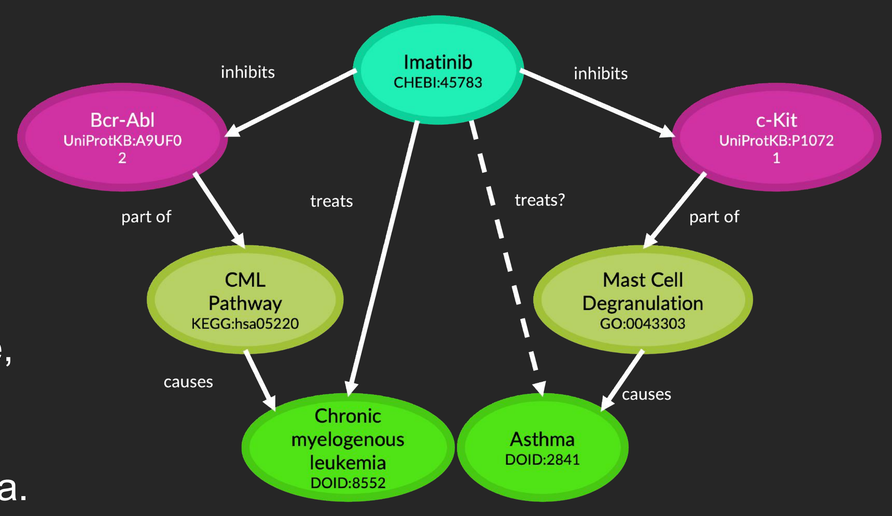
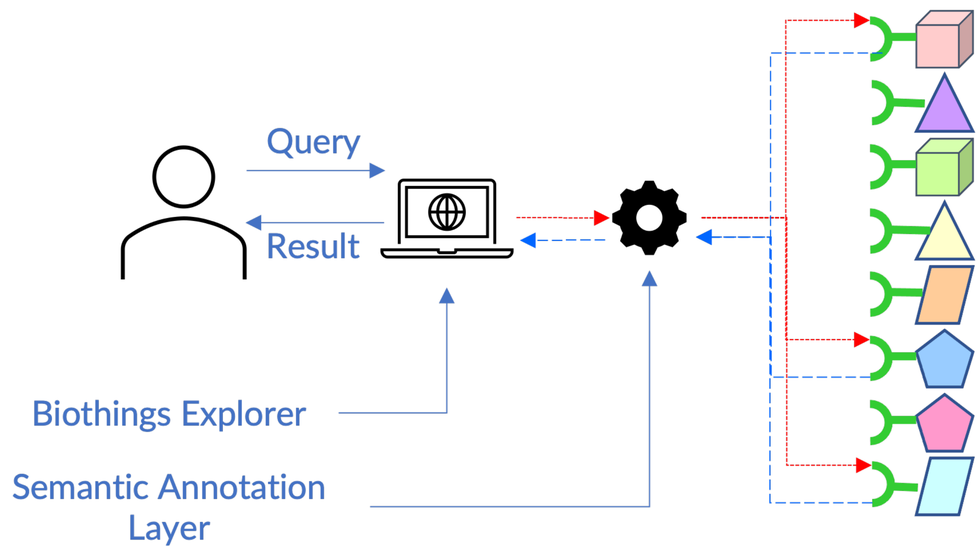
Biomedical Knowledge Graphs

Knowledge graphs (KGs), heterogenous structures of concepts and relationships, are a well-suited structure for representing biomedical knowledge. Algorithms for operating on them are well-established in all kinds of software packages. As seen in the diagram, KGs lend themselves to finding complex relationships to explain existing knowledge, or inferring the possibility of a connection which may warrant further study. Based on known relationships (far right), Imatinib is currently studied as a potential treatment for asthma.



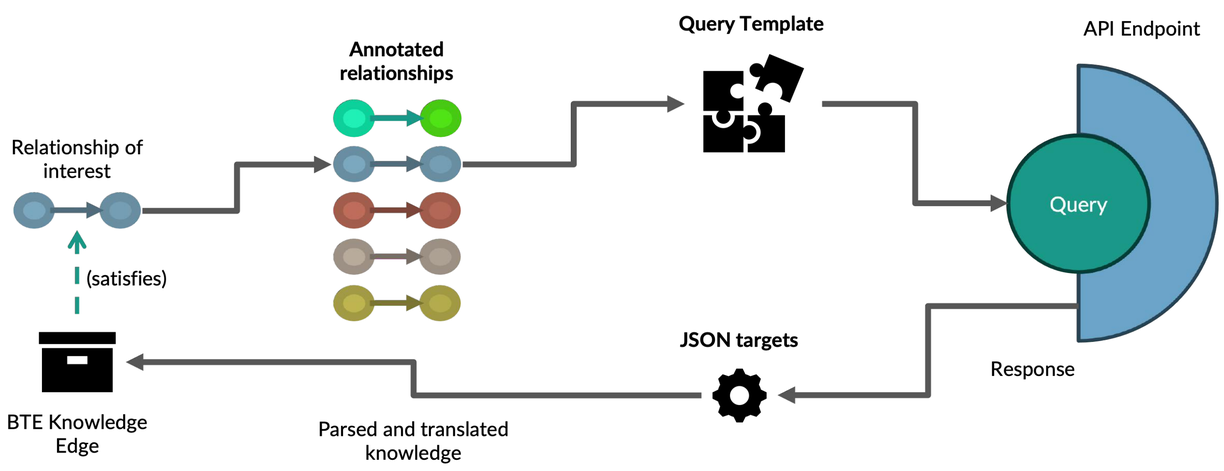
Distributed knowledge: Data Federation

Bringing together disparate sources of knowledge without requiring laborious consolidation efforts can be accomplished by data federation, where individual knowledge providers are able to join on at will, with a service providing a means of centralized access and integration. BioThings Explorer (BTE) acts as this service, making use of a translation layer (by way of semantic annotation) to manage API interactions and knowledge interoperability.



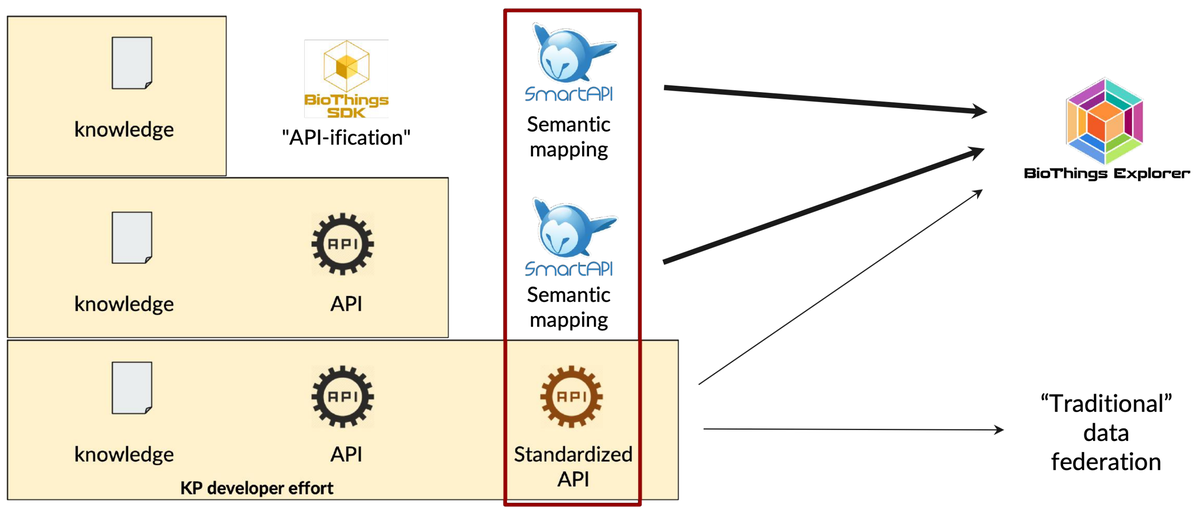
Semantic Annotation as a translator

By mapping knowledge an API can provide to the BioLink Model, it becomes possible to group numerous ID namespaces under the same semantic types, with standardized meanings between different sources. This mapping is done in an annotation file, which also provides a specification of the API using the OpenAPI standard. Different API structures can then be queried for interoperable information.



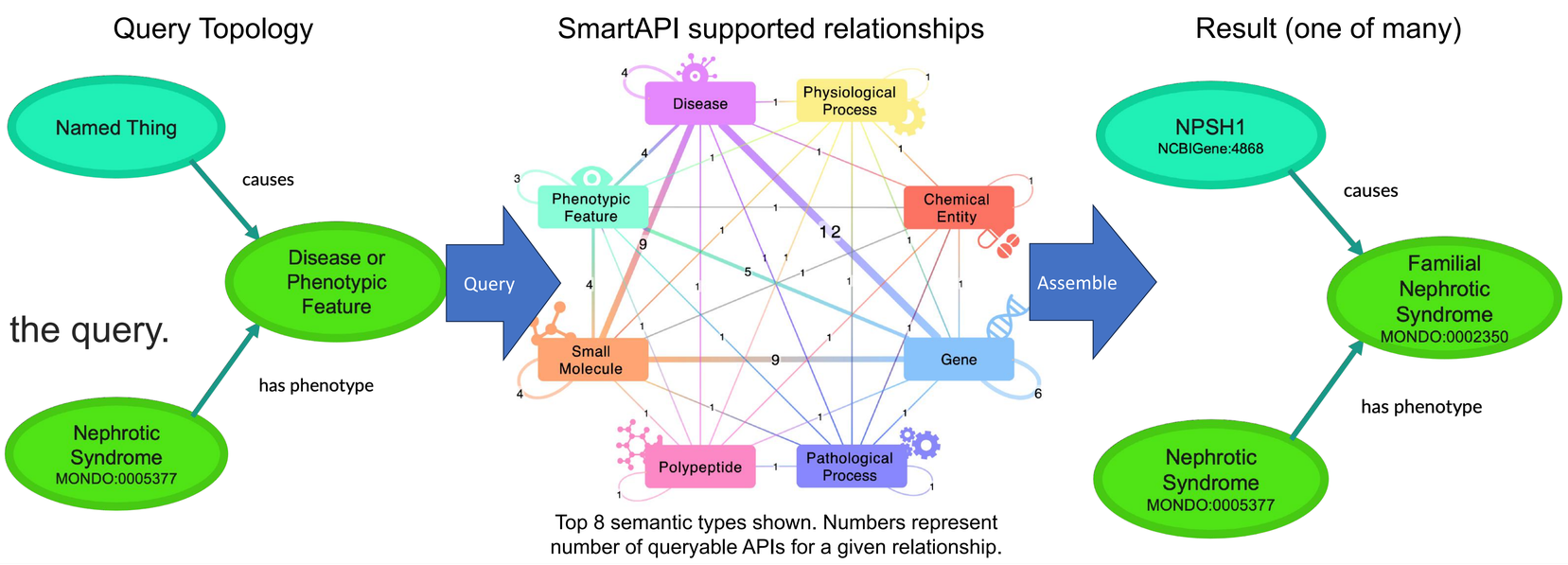
Lowering the barrier for federation

By using an annotation file to federate, knowledge providers are not required to make code changes to be able to interoperate with BioThings Explorer. The knowledge provider may operate as independently from the ecosystem as they see fit. If the provider does not have an API, one can easily be made using the BioThings SDK. Annotations are housed in the SmartAPI registry, which BioThings Explorer references.



Seamless querying

BioThings Explorer may be queried with any acyclic graph, and will assemble matching results from the knowledge available across all providers which serve relationships contained in the query. Answers may come back using edges from several APIs.



Top 8 semantic types shown. Numbers represent number of queryable APIs for a given relationship.

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Further Reading